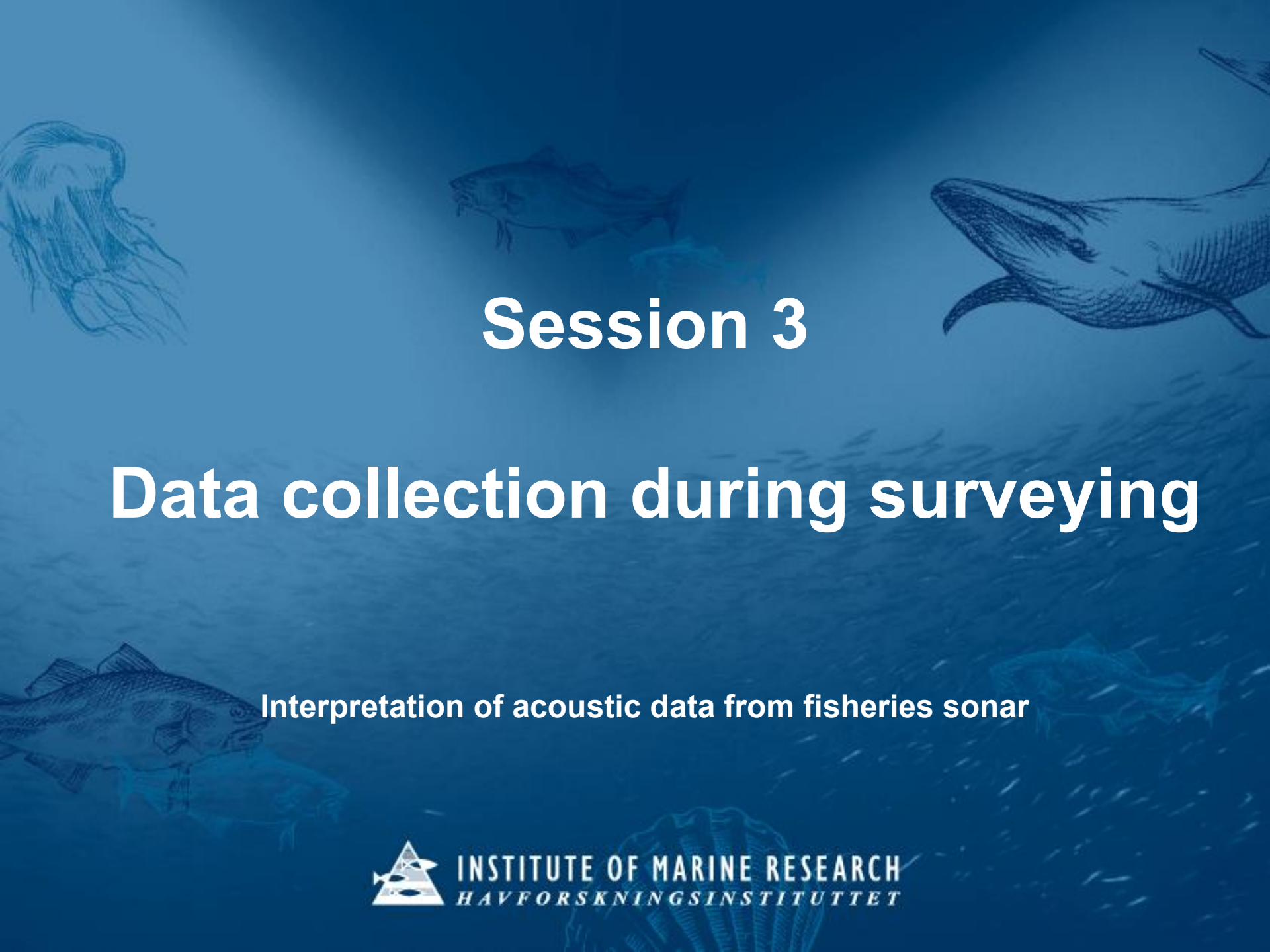




Session 3

Data collection during surveying

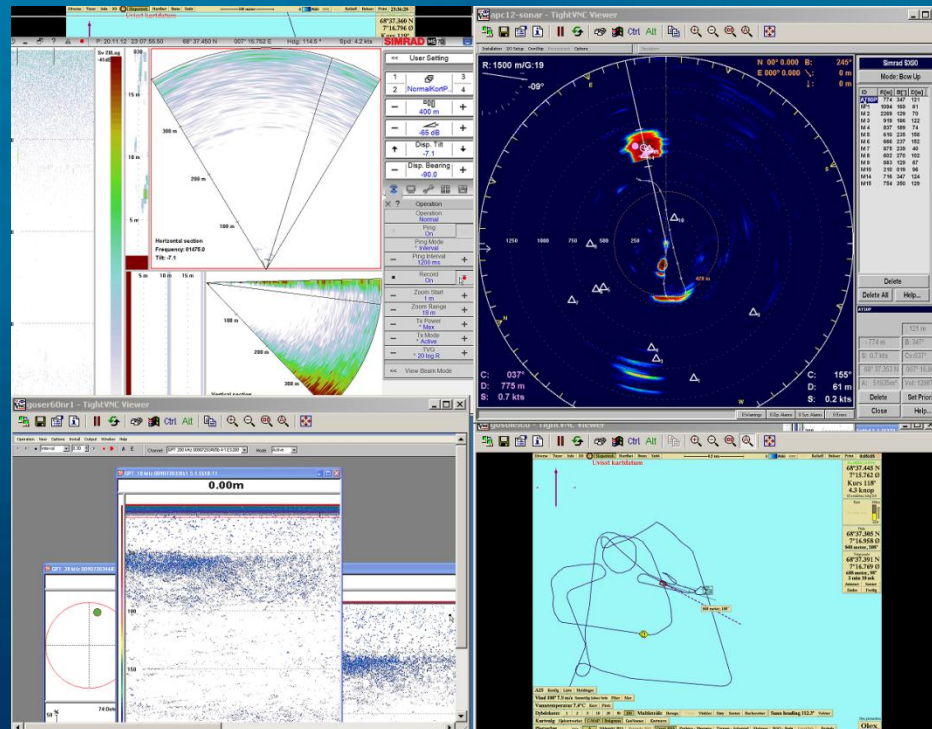
Interpretation of acoustic data from fisheries sonar



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Before data collection

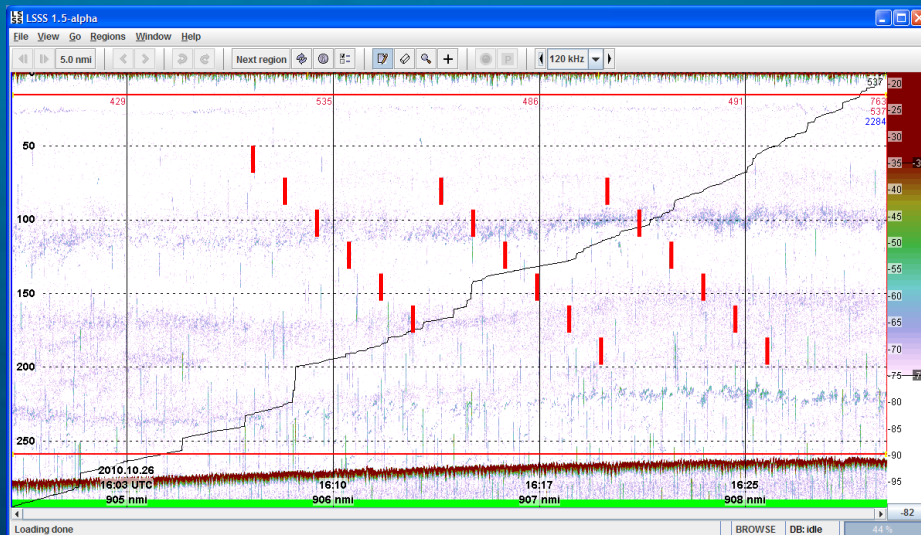
- Synchronization of all clocks to UTC time, **up to the second**
- sonar PC, echo sounder PC, chart PC
 - All files names includes the PC time, and should be the same to facilitate post-processing



Before data collection

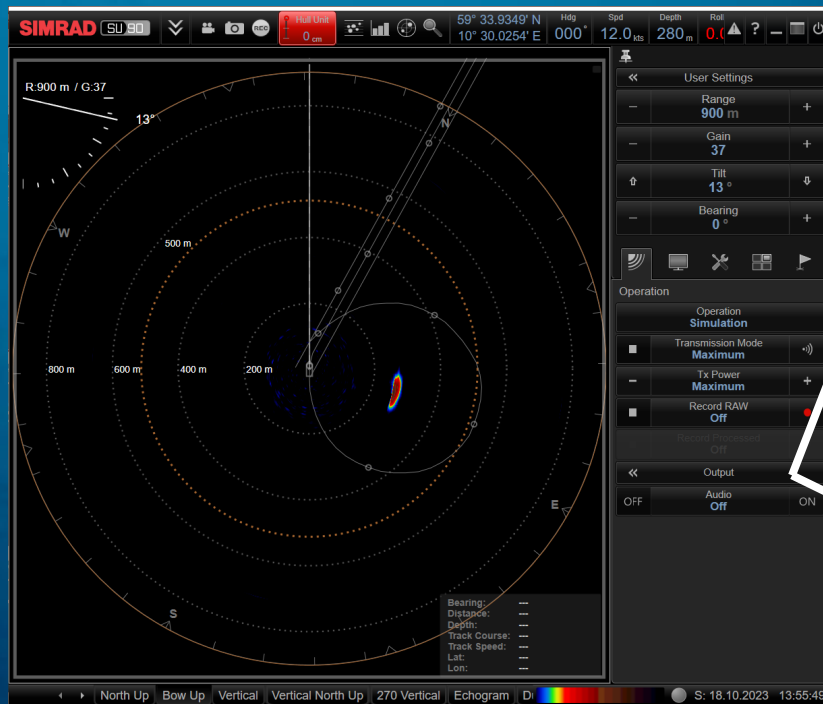
Synchronization of sonar and echo sounder

- Avoid acoustic interference
- Normally echo sounder is **Master** and sonar is **Slave**
- Average ping rate, 1 ping per second
- Instrument personnel knows how to sync



Before data collection

In Windows 10 software, if activated, you can choose to record .RAW or record netCDF files (named "Processed Data")



RAW file options

File Setup

I/O Setup

Processed Data to File

Marker Output

Relay Output

Screen Recording

General

Current Output Folder: C:\Projects 2022\Maitokki paper sonar echo\Simulator fo [Browse...]

File Name

File Name Prefix: []

File Size

Maximum File Size: 100 [MB]

Current File Size: 0 [B]

OK Cancel Apply

netCDF file options

File Setup

I/O Setup

Processed Data to File

Marker Output

Relay Output

Screen Recording

Installed Outputs

New

Processed Data to File

Output: Sonar-netCDF4

Current Output Folder: D:\Recording [Browse...]

File Name

File Name Prefix: []

File Size

Maximum File Size: 100 [MB]

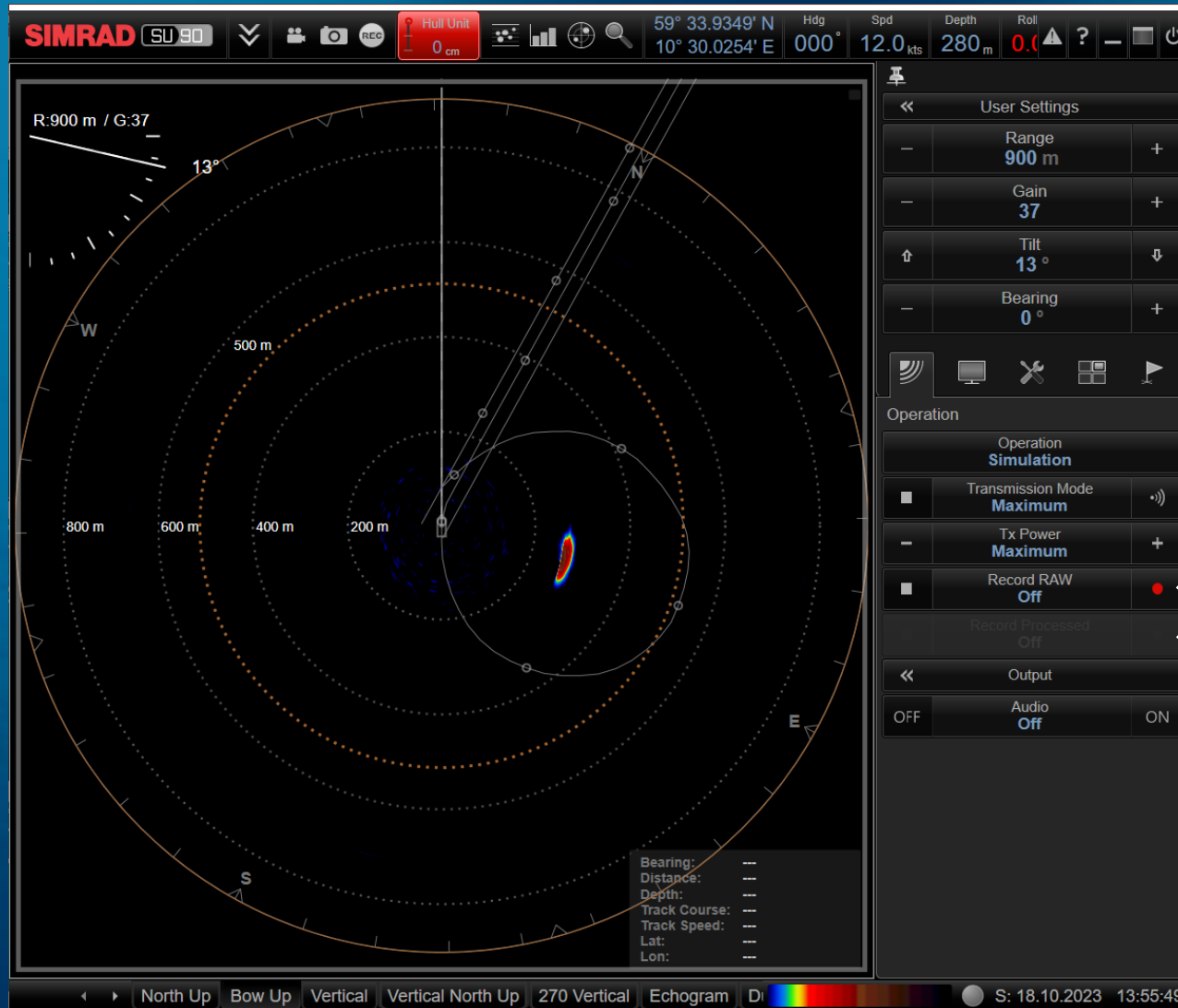
Min. free disk space: 5120 [MB]

Add

OK Cancel Apply

Before data collection

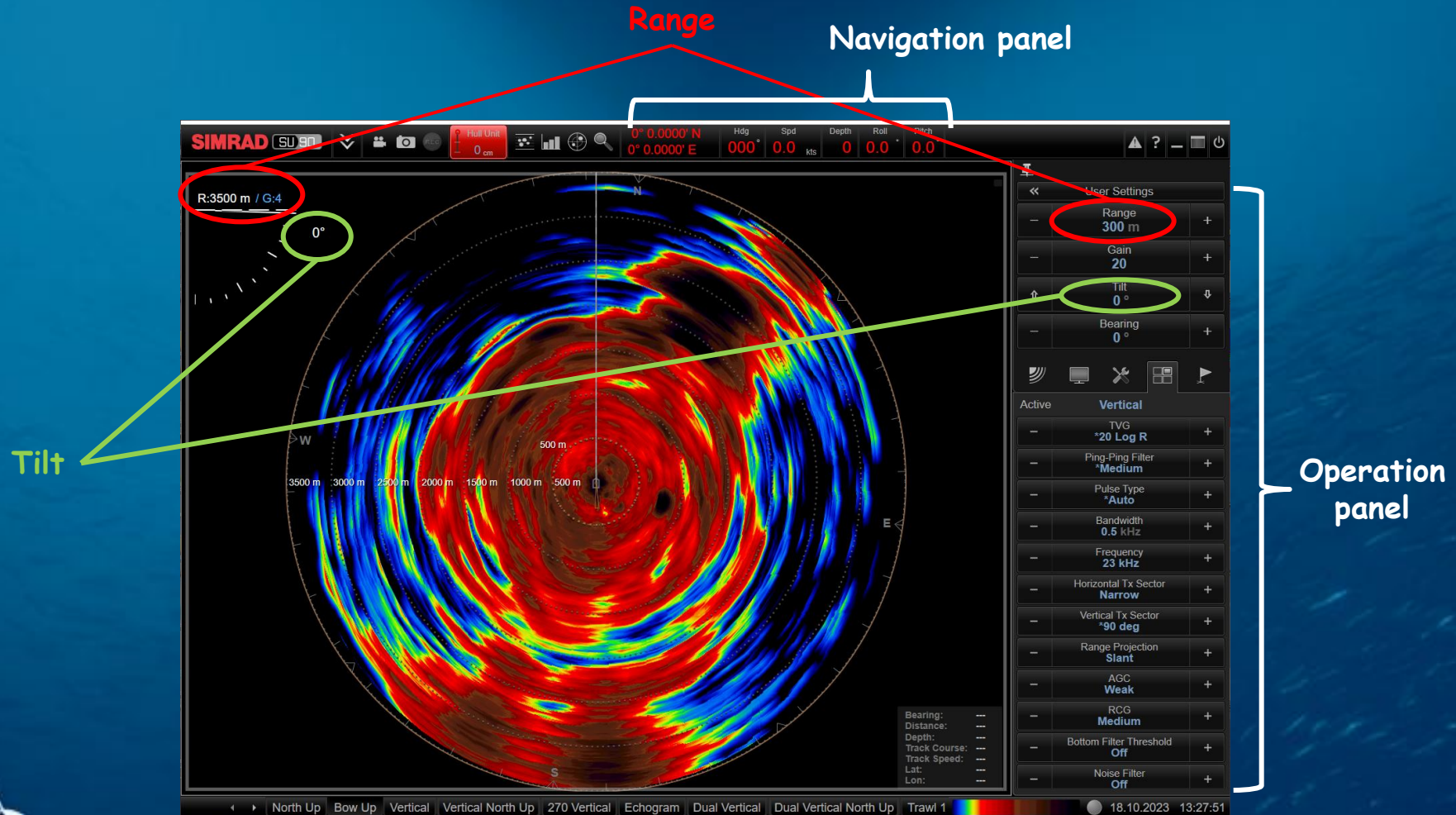
- Start storing data



click for .raw
recording

click for .netCDF
recording

Typical sonar display with Bow up mode (W10)



Data collection

- Continuous recording during whole survey
- Sonar data output rate, ca. 4 GB/hour, 110 GB/day ($f(\text{frequency}, \text{sonar range})$)
- Need continuous watch during data collection same as echo sounder
- Replay collected data in LSSS in a regular basis, to ensure data has no errors
- Create log book (or use Tokt logger), to register periods with schools detections or significant events. Include time (UTC), size of schools (small, medium, large), range and tilt

